Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 24-03-2026

Relational Inventory Control & Stock Tracking System

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

Presented By:

P.RAHUL– 26295– 23UECS0430

Courses Handling Faculty:

Dr.KAVIARASAN/ASST PROFFESSOR

INTRODUCTION

- Managing inventory manually often leads to errors, data loss, and inefficiency in tracking stock levels.
- Businesses face challenges in maintaining accurate records of products, sales, and stock updates.
- This project is important as it automates inventory management using a relational database system.
- It helps reduce human errors and improves accuracy in stock tracking and reporting.
- The system has real-world relevance in retail shops, warehouses, and small businesses.
- It ensures better decision-making through real-time data on product availability and sales.
- The system includes features like product addition, stock updates, and sales management.
- Overall, it provides a structured and efficient way to manage inventory using database relationships.

PROBLEM STATEMENT

- Inventory management is important for businesses to track products and materials.
- Many small businesses still use manual records or spreadsheets.
- This leads to errors, stock loss, and poor tracking.
- The proposed system provides a relational database-based inventory control system.
- It helps monitor stock levels, product details, and transactions.
- The system improves accuracy and efficiency in stock management.

OBJECTIVE(S)

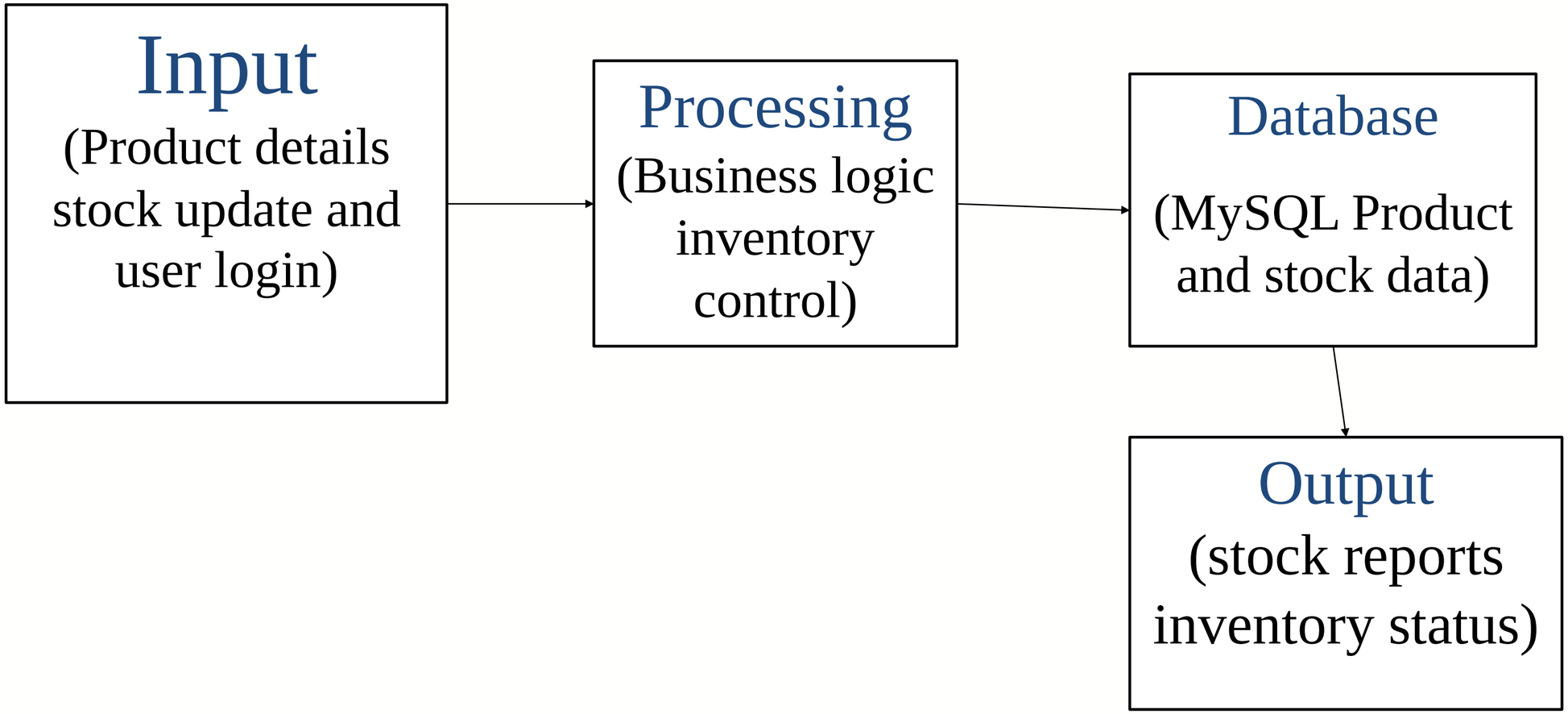
1. To design and develop a relational inventory management system.

2. To implement stock tracking and reporting features.

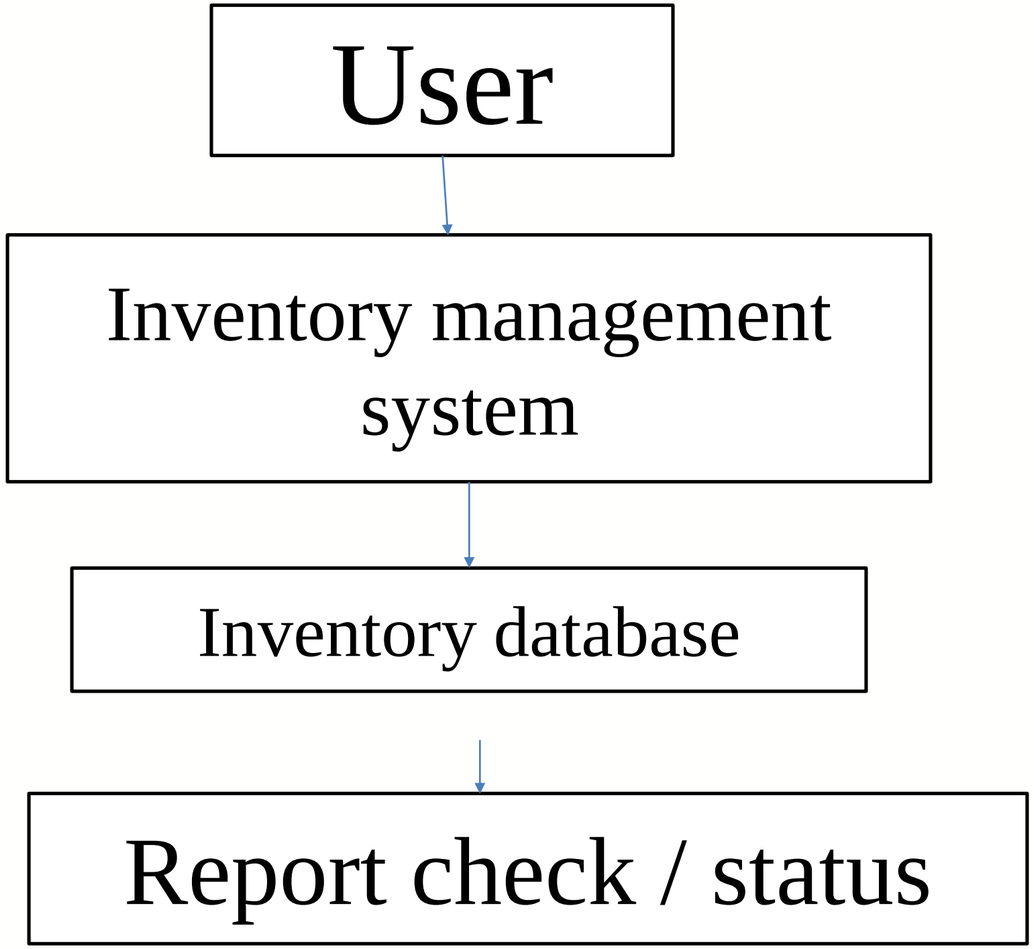
PROPOSED SYSTEM

- Digital inventory tracking system
- Uses relational database for structured data
- Tracks product quantity and transactions
- Provides real-time stock updates
- Generates reports for inventory analysis
- Reduces manual errors
- Improves stock monitoring
- Easy product management

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW


User
Inventory management system
Inventory database
Report check / status

TECHNOLOGIES USED

- Frontend: HTML, CSS, JavaScript
- Backend: Java (Spring Boot)
- Database: MySQL
- Tools: VS Code,
- Hardware/Software: Computer with Internet

MODULE DESCRIPTION

Module 1: User Management – Handles login and authentication

Module 2: Inventory Management – Adds and updates products

Module 3: Stock Tracking – Tracks stock in and stock out

Module 4: Reporting – Generates inventory reports

IMPLEMENTATION SCREENSHOTS

Home Page – Displays system dashboard

Login Page – Secure user login

Inventory Page – Manage products and stock

IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8(A)
9

Sales Entry

Product ID

Quantity

Record Sale

Product List

ID	Name	Category	Price	Quantity

IMPLEMENTATION SCREENSHOTS

Inventory Control System

Add Product View Products Sales

Dashboard

Total Products: 100

Low Stock Items: 5

Total Sales Today: ₹5000

IMPLEMENTATION SCREENSHOTS

Add Product

Product Name:

Category:

Price:

Quantity:

Add Product

ADVANTAGES & LIMITATIONS

- Easy inventory monitoring
- Reduces manual work
- Accurate stock tracking
- Faster reporting
- Requires internet and system access
- Initial setup required
- Limited to basic inventory functions

Thank you